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# PERFORMANCE SPECIFICATION

# SAMPLING AND GAGING KIT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers a portable petroleum sampling and gaging kit. The equipment is used to perform American Petroleum Institute (API) and American Society for Testing and Materials (ASTM) petroleum sampling procedures, calculating API gravity, and ascertaining the amount of water in the storage tank.

## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

Comments, suggestions, or questions on this document should be addressed to U.S. Army RDECOM, Tank Automotive Research, Development and Engineering Center, ATTN: RDTA-SIE-ES-PLE-SI MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or sent by email to usarmy.detroit.rdecom.mbx.tardec-standardization@mail.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil

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2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130 - Identification Marking of U.S. Military Property

(Copies of this document are available online at http://quicksearch.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN PETROLEUM INSTITUTE (API)

| | | |
|---|---|---|
| MPMS 3.1A | - | Manual of Petroleum Measurement Standards Chapter 3.1A – Third Edition |
| MPMS 7 | - | Manual of Petroleum Measurement Standards Chapter 7 - Temperature Determination – First Edition |

(Copies of these documents are available from www.api.org or www.ihs.com).

ASTM INTERNATIONAL

| | | |
|---|---|---|
| ASTM D1250 | - | Petroleum Measurement Tables, American Edition, Tables 5 and 6 |
| ASTM D1298 | - | Standard Practice for Density, Relative Density (Specific Gravity), or API Gravity of Crude Petroleum And Liquid Petroleum Products by Hydrometer Method |
| ASTM D4057 | - | Standard Practice for Manual Sampling of Petroleum and Petroleum Products |
| ASTM D7777 | - | Standard Test Method for Density, Relative Density, or API Gravity of Liquid Petroleum by Portable Digital Density Meter |

(Copies of these documents are available from www.astm.org).

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE-AMS-STD-595 - Colors Used in Government Procurement

(Copies of this document can be found at www.sae.org, or SAE International Customer Service, 400 Commonwealth Drive, Warrendale, PA 15096).



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2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

3.1 Description. The sampling and gaging test kit shall consist of the components specified herein and be fitted in a portable case.

3.2 First article. When specified (see 6.2), a sample test kit shall be subjected to first article inspection in accordance with 4.2.

3.3 Materials. The contractor shall select materials capable of meeting all the operational and environmental requirements specified herein. Test kit components shall be resistant to fuel and fuel esters, ultraviolet radiation and ozone deterioration. Unbreakable, or break resistant, materials shall be used as practicable.

3.3.1 Material deterioration and control. The kit and its components shall be fabricated from compatible materials, inherently corrosion and deterioration resistant or treated to provide protection against the forms of corrosion and deterioration to which they are susceptible.

3.3.2 Dissimilar metals. Dissimilar materials shall be electrically isolated from one another to minimize or prevent galvanic corrosion.

3.3.3 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs. Used, rebuilt, or remanufactured components, pieces and parts shall not be incorporated in the test kit.

3.4 Operating parameters.

3.4.1 Major components. The kit shall be provided with all components and accessories required to perform initial sampling and not less than 20 (each) of the listed functions, on individual samples, without replenishment. The kit shall include instruction plate(s), or booklet(s) on how to perform the procedures for testing of fuel. The kit shall also include a table or card for correcting API gravity readings at any temperature to API gravity readings at 60 °F and the reverse. The sampling and gaging accomplished with this set shall be as follows:

a. API gravity per ASTM D1298 or ASTM D7777
b. Fuel sampling per ASTM D4057 (beaker sampling - 1L weighted beaker stainless steel; All Levels Sample; volume = 0.8±0.25 L)
c. Fuel gaging per API MPMS 3.1A (Chapter 3, Section 1a)
d. Fuel temperature per API MPMS 7 (Chapter 7)
e. API gravity temperature conversion per ASTM D1250 (Tables 5 and 6)


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3.4.2 Case design. The kit components shall be contained in a case. Components shall be securely restrained, and adequately cushioned to prevent damage during transit.

3.4.2.1 Envelope. The overall dimensions of the case, in storage or transit configuration, shall not be greater than 46 cm (18.00 inches) long × 46 cm (18.00 inches) deep × 30 cm (12 inches) high.

3.4.2.2 Weight. The weight of the kit shall not exceed 14.0 kg (31 pounds).

3.4.2.3 Handles. The design of the case shall incorporate a strap, or other suitable means, for grasping, handling, and carrying by one person wearing appropriate protective clothing required for fuel sampling.

3.5 Test kit loading diagram. A diagram shall be provided to identify equipment placement in the case. The loading diagram shall be permanently affixed to the inside of the kit's cover to easily identify storage locations. The loading diagram shall be legible for the life of the kit.

3.6 Environmental requirements.

3.6.1 Climatic. The kit shall be capable of performing in any ambient temperature from -25°C (77°F) to 50°C (122°F). When placed in storage, the test kit shall not be damaged in any ambient temperature from -33°C (-27.4°F) to 63°C (145.4°F) (see 4.5.2).

3.6.2 Fungus resistance. Components susceptible to fungus and moisture damage shall be coated for protection.

3.6.3 Water resistance. The case shall not allow any water leakage to its interior.

3.6.4 Transportability. The kit shall be capable of withstanding the vibration stress and shock encountered when transported by all modes of transportation.

3.7 Personnel environment.

3.7.1 Human factors. Compartment access and grasp areas shall reflect compatibility with the Clothing and Personal Equipment (C/PE) of personnel using and maintaining the kit, under the environmental conditions specified herein. Grasp areas shall be located to provide easy accessibility, and shall not interfere with installation, removal, operation, or maintenance.

3.7.2 Safety. Adequate safeguards shall be incorporated into the kit and its components so that personnel shall not be exposed to concentrations of toxic or corrosive substances. Except where functionally required, exposed surfaces shall be free from burrs, sharp edges and corners, or other features that present a personnel safety hazard; or shall be shielded against contact.

3.8 Treatment and painting. Unless otherwise specified (see 6.2), the case shall be cleaned, treated, and painted in accordance with the manufacturer's standard commercial practice and MIL-C-46168 shall be used as a guide for coating. The color shall be Army green #383, color #34094 of SAE-AMS-STD-595.

3.9 Labeling. Labels, legends, placards, signs, markings, or a combination of these shall be provided for personnel to identify, interpret, follow procedures, and to avoid hazards. Label characteristics shall be consistent with the accuracy of identification required, time available for


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recognition and appropriate response, distance at which the labels must be read, and with label design within and between units. Labels shall be permanent, clear, distinct; and have a high contrast, foreground to background. The weight shall be shown on a durable, corrosion resistant plate permanently affixed to the exterior of the case. The plate shall be legible for the life of the kit.

3.9.1 Identification markings. The kit shall be identified to include item nomenclature, NSN, manufacturer's part number, manufacturer's date, Commercial and Government Entity (CAGE) code, contract number and serial number, following criteria prescribed in MIL-STD-130.

## 4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. When a first article inspection is required, it shall be performed on one complete test kit. The inspection shall include the examination of 4.4 and the testing of 4.5.

4.3 Conformance inspection. Conformance inspection shall include the examination.

4.4 Examination. Each test kit shall be physically examined for the deficiencies listed in Table I. The presence of one or more defects shall be cause for rejection.

4.5 Tests. The tests shall be conducted in the following order: 4.5.1.1 through 4.5.1.5 (Performance), 4.5.3.1 (Transportability), 4.5.3.3 (Shock), 4.5.2.5 (Water Resistance), 4.5.3.2 (Vibration), 4.5.2.1 through 4.5.2.4 (High/Low Temp - Storage/Operational).

4.5.1 Performance. Unless otherwise specified, performance testing shall be conducted at room temperature 20±2 °C (68°F).

4.5.1.1 Fuel temperature. The test shall be conducted in accordance with API-MPMS Chapter 7 - Section 1. Missing components or supplies required for the test, failure to complete, or to meet applicable criteria shall constitute failure of this test.

4.5.1.2 Fuel sampling. Fuel sampling shall be conducted in accordance with ASTM D4057, using the weighted beaker procedure. Missing components or supplies required for the test, failure to complete, or inability to collect a sample of the specified volume (section 3.4.1b) shall constitute failure of this test.



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TABLE I. Examination.

| No. | Examination Description | Reqt Para | Method |
|---|---|---|---|
| 101 | Components and consumables not as specified | 3.4.1 | Visual/Doc. |
| 102 | Case not as specified | 3.4.2 & 3.4.2.1 | Visual/Doc./SIE |
| 103 | Component placement not as specified | 3.5 | Visual/Doc. |
| 104 | Weight not as specified | 3.4.2.2 | Visual/Doc./SIE |
| 105 | Protection devices not as specified | 3.7.2 | Visual/Doc. |
| 106 | Accessibility not as specified | 3.7.1 | Visual/Doc./SIE |
| 107 | Labeling not as specified | 3.9 | Visual/Doc. |
| 108 | Identification marking not as specified | 3.9.1 | Visual/Doc. |
| 109 | Materials not as specified | 3.3 & 3.6.2 | Visual/Doc. |
| 110 | Deterioration prevention and control not as specified | 3.3.1 | Visual/Doc. |
| 111 | Dissimilar metals not protected against galvanic corrosion as specified | 3.3.2 | Visual/Doc. |
| 112 | Used, rebuilt or remanufactured parts utilized not as specified | 3.3.3 | Visual/Doc. |
| 113 | Treatment/painting not as specified | 3.8 | Visual/Doc. |

Doc. = Documentation, SIE = Standard Inspection Equipment

4.5.1.3 Relative density. The test shall be conducted in accordance with ASTM D1298 or ASTM D7777 with a known test sample. Missing components or supplies required for the test or failure to complete shall constitute failure of this test.

4.5.1.4 Fuel gaging. The test shall be conducted in accordance with API-MPMS 3.1A. Missing components or supplies required for the test, failure to complete, or to meet applicable criteria shall constitute failure of this test.

4.5.1.5 Equipment and material quantities. The types and quantities of the equipment and materials used to conduct the tests in 4.5.1.1 through 4.5.1.4 shall be extrapolated to determine the capability of the kit to perform initial sampling and at least 20 repetitions of the required functions (see 3.4.1).

4.5.2 Environmental.

4.5.2.1 High temperature - storage. The test kit, with case closed, shall be subjected to a high temperature environment of 63±1 °C (145.4°F) for 4 hours, after an initial period of 6 hours at 49±1 °C (120.2 °F). Raising and lowering of the temperature shall be accomplished within a period of 1 hour, for a total cycle time of 12 hours. A minimum of seven cycles (84 hours) is required. At completion of the last cycle, adjust chamber air temperature to 20±1 °C (68 °F) and maintain until temperature stabilization of the test kit has been achieved. Conduct the test procedures outlined in 4.5.1.3 and 4.5.1.4. In addition to the conformance criteria cited therein, the kit case and packing material shall be examined for deterioration, brittleness, or cracking. Functional failure of the kit components, case, or packing material, or inability to perform tests, or satisfy test precision criteria shall constitute failure of this test.



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4.5.2.2 High temperature - operational. The test kit, with the case open, shall be subjected to a high temperature of 50±1 °C (122 °F) for a period of 4 hours. Conduct the test procedures outlined in 4.5.1.3 and 4.5.1.4 while the temperature is maintained at 50±1 °C (122 °F). In addition to the conformance criteria cited therein, functional failure of the kit components, inability to perform the tests or satisfy test precision criteria shall also constitute failure of this test.

4.5.2.3 Low temperature - storage. The test kit, with the case closed, shall be subjected to a low temperature environment of -33±1 °C (-27.4 °F) for a period of 48 hours. At the end of this exposure, adjust the chamber air temperature to 20±1 °C (68 °F) and maintain until temperature stabilization of test kit has been achieved. Conduct the test procedures cited in 4.5.1.3 and 4.5.1.4. In addition to the conformance criteria cited therein, the kit case and packing material shall also be examined for deterioration, brittleness, or cracking. Functional failure of the kit components, case, or packing material, or inability to perform tests, or satisfy test precision criteria shall constitute failure of this test.

4.5.2.4 Low temperature - operational. The test kit, with case open, shall be subjected to a low temperature environment of -25±1 °C (-13 °F) for a period of 4 hours. At the end of this exposure, while still at -25±1 °C (-13 °F), conduct the test procedures cited in 4.5.1.3 and 4.5.1.4. In addition to the conformance criteria cited therein, functional failure of the kit components, inability to perform the tests or satisfy test precision criteria shall also constitute failure of this test.

4.5.2.5 Water resistance. For the water resistance test, the kit shall be in its transit or storage configuration. A water spray shall be directed through nozzles to cover the parting line between sections of the case, hinge(s), and the carrying handle(s). The spray nozzle shall give uniform water distribution and have a flow capacity of not less than 4 liters/minute (1 gallon/minute), at a 207 to 276 kPa (30-40 psi) water line pressure, and water temperature between 4±1 °C (39.2 °F) and 24±1 °C (75.2 °F). Fogging nozzles shall not be used. The spray shall be angled to each of the surfaces so water strikes at an angle between 40° and 90° to the surface. Spray time on each case surface shall not be less than 3 minutes continuous. At the conclusion of the test, the case shall be examined visually for water leakage to its interior. Water leakage shall constitute failure of the test.

4.5.3 Transportability.

4.5.3.1 Transportability. The kit shall be put in the transport mode and carried by one person for a distance of 50 meters (164 ft). The contents shall be examined for damaged items or items not secured. All items shall then be removed from their secured positions. Components shall be re-packed, using the loading diagram as a guide, the case returned to transport mode, and carried by one person for a distance of 50 meters. Upon completion, the kit shall be examined for damage to the case and components. The inability to successfully carry the kit for the specified distance, inability to re-pack components per the loading diagram, changes in configuration, or difficulty in re-packing resulting from improper fit, shall constitute failure of this test. Any permanent deformation, cracks, or damaged items from any cause shall also constitute failure of this test.

4.5.3.2 Vibration. The kit shall endure 40 minutes of vibration in each of 3 axes. The test bed shall produce a 2.5 cm (1.0 inch) double amplitude, orbital path, at 5 Hz minimum. Upon completion of this test, each kit shall be examined for damage to the case and each component and accessory. Damage or breakage of the case or any component of the kit shall constitute failure of


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this test.

4.5.3.3 Shock. The kit shall be dropped onto a 20 cm (8.00 inch) thick bed of dry sand, from a height of 122 cm (48.00 inches). The kit shall be dropped on the two edges along the length of the bottom and the two edges along the width of the bottom. For edgewise drops, the striking edge of the kit shall be parallel to the dropping surface and the kit's center of gravity shall be directly above the striking edge, at the instant of release. Drop tests shall be conducted at -32±3 °C (-25.6 °F) and 49±3 °C (120.2 °F). The kit temperature shall be at the specified ambient temperature prior to the drop tests. The kit shall be subjected to a total of four drops at each temperature specified. All drops shall be made so that the test kit falls freely through the distance of 122 cm (48 inches). Upon completion of the drops at each temperature, the kit shall be examined for damage to the case and each component and accessory. Any permanent deformation, cracks, or damaged items from any cause shall also constitute failure of this test. Any item previously restrained that has detached from its storage compartment shall also constitute failure of this test.

## 5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by Inventory Control Point’s packaging activities within the Military Service or Defense Agency, or within the military service’s system commands. Packaging data retrieval is available from the managing Military Department’s or Defense Agency’s automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

## 6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The kit is intended for use in the field to provide an indication of product type or check for commingling of fuels. The kit covered by this performance specification is military unique because it is used in the brigade forward supply section of all divisions and has strategic mobility. In addition, the kit must be able to perform under adverse conditions and withstand drops above 1 m. Commercial equipment is not designed to withstand these harsh conditions.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When first article is required (see 3.2).
c. When special cleaning, treatment, or color is required (see 3.8).
d. Packaging requirements (see 5.1).

6.3 First article inspection. When a first article inspection is required, the item should be a first article production unit. The first article should consist of one complete test kit. The contracting officer should include specific instructions in acquisition documents regarding


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arrangements for examinations, tests, and approval of the first article test results and disposition of the first article.

6.4 Data requirements. The contracting officer should include requirements for such data as parts lists, consumable items lists (including source data), etc., to be furnished with each test kit.

6.5 Definitions.

6.5.1 Dissimilar metals. For the purposes of this specification, metals are dissimilar when two metal specimens in contact with each other promote accelerated galvanic corrosion.

6.5.2 Inspection. Inspection, as used in this specification, is defined as both examination such as visual or auditory investigation without the use of special laboratory appliances or procedures of the unit.

6.5.3 Recycled material. For the purpose of this requirement, recycled materials are those that have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials.

6.5.4 Transport/storage mode. For the purpose of this specification, transport mode is when the kit is closed and locked and handles are prepared for transporting/storage.

6.6 Subject term (key word) listing.

API gravity
Contaminants
Fuel
Temperature
Water indicator

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.



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